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PAPER OF CSE-2025 - PYTHON TEST9 EXAM

Q1. Power Grid Surge History

An electrical engineer is analyzing power grid data, where each reading is classified as '**H**' (**High/Surge**) or '**L**' (**Low/Normal**). A long consecutive surge period can indicate a fault.

Calculate the **maximum continuous duration of power surges** (consecutive 'H' characters) from the input data.

Input Format:

A string 'data' where 'H' and 'L' represent the power status history.

Output Format:

Print the maximum continuous duration of power surges (consecutive 'H' characters) from the input data.

Sample Input:

```
"LHHHLHHHHLLH"
```

Sample Output:

```
4
```

ANSWER:

Test case 1 :

Input:

```
LHHHLHHHHLLH
```

Output:

```
4
```

Test case 2 :

Input:

```
LLLLL
```

Output:

```
0
```

Test case 3 :

Input:

```
HHH
```

Output:

3

Test case 4 :

Input:

HLHLH

Output:

1

Test case 5 :

Input:

Output:

0

Test case 6 :

Input:

H

Output:

1

Q2. City Traffic Jam Monitor

The city's transport department monitors highway conditions minute-by-minute, logging the state as either '**F**' (**Free Flowing**) or '**S**' (**Slow/Stuck**).

They want to identify the maximum duration of a single, continuous traffic jam. You must analyze the traffic log's string 'traffic_log' and determine the **longest sequence of consecutive 'S'** (Slow/Stuck) entries.

Input Format:

Given a sequence of single-character traffic states in string 'traffic_log'.

Output Format:

Find and Print the integer value representing the **longest sequence of consecutive 'S'** (Slow/Stuck) entries

Sample Input:

"FSSSFSSSSSF"

Sample Output:

5

ANSWER:

Test case 1 :

Input:

FSSSFFSSSSSF

Output:

5

Test case 2 :

Input:

FFFFFFF

Output:

0

Test case 3 :

Input:

SSS

Output:

3

Test case 4 :

Input:

S

Output:

1

Test case 5 :

Input:

SFFSSSFFFFS

Output:

3

Test case 6 :

Input:

Output:

0

Q3. Student Exam Pass Chain

A teacher records the pass/fail results for a large group of students in a single sequence: **'P' (Pass)** or **'F' (Fail)**.

They are looking for the longest chain of students who passed their exam consecutively.

Given the result sequence 'result', calculate the length of the **longest consecutive chain of 'P' (Pass)** results.

Input Format:

A string 'result', containing only 'P' and 'F' characters.

Output Format:

Find and print the integer value representing the length of the **longest consecutive chain of 'P'** (Pass) results.

Sample Input:

"FFPPFP"

Sample Output:

3

ANSWER:

Test case 1 :

Input:

FFPPFP

Output:

3

Test case 2 :

Input:

FFFFFFF

Output:

0

Test case 3 :

Input:

PPP

Output:

3

Test case 4 :

Input:

P

Output:

1

Test case 5 :

Input:

FPFPFPFFF

Output:

2

Test case 6 :

Input:

Output:

0

Q4. Balloon Line Safety Check

A row of balloons is tied in a straight line.

Each balloon has a **gas level** inside it.

The safety checker looks at every **3 balloons next to each other**.

If the **total gas level** of those 3 balloons is **more than 60**, the **middle balloon** bursts and disappears.

This checking keeps happening until no group of 3 balloons has gas above 60.

Your task is to return the **final list of balloon gas levels** after everything becomes safe.

Rules :

- Look at **3 balloons next to each other**.
- If the total is **more than 60**, remove the **middle** one.
- Keep doing this until **no more** groups of 3 are above 60.
- Return the final list.

Input Format:

A list of integers showing the gas in each balloon.

Output Format:

A list of integers showing the balloons left after safety check.

Sample Test Case 1:

Input:

[25, 20, 30]

Output:

[25, 30]

Sample Test Case 2:

Input:

[10, 45, 20, 15]

Output:

[10, 20, 15]

ANSWER:

Test case 1 :

Input:

[25, 20, 30]

Output:

[25, 30]

Test case 2 :

Input:

[10, 20, 15, 10]

Output:

[10, 20, 15, 10]

Test case 3 :

Input:

[50, 15, 10, 20, 15]

Output:

[50, 15]

Test case 4 :

Input:

[15, 15, 15, 15, 15]

Output:

[15, 15, 15, 15, 15]

Test case 5 :

Input:

[30, 20, 15, 10]

Output:

[30, 15, 10]

Test case 6 :

Input:

[40, 25, 10]

Output:

[40, 10]

Q5. Car Speed Line Check

A line of toy cars is moving on a small track.

Each car has a **speed number**.

The track controller checks every **3 cars next to each other**.

If the **total speed** of those 3 cars is **more than 60**, the **middle car** is removed from the track for safety.

This checking continues until **no group of 3 cars** has a total speed above 60.

Your task is to return the **final safe list** of car speeds.

Rules :

- Look at every group of **3 cars**.
- If their total speed is **more than 60**, remove the **middle car**.
- Keep repeating until the list is safe.
- Finally, return the safe list of speeds.

Input Format:

A list of integers showing the speed of each car.

Output Format:

A list of integers showing the final safe speeds of the cars.

Sample Test Case 1:

Input:

[15, 30, 20]

Output:

[15, 20]

Sample Test Case 2:

Input:

[40, 10, 25, 20]

Output:

[40, 25, 20]

ANSWER:

Test case 1 :

Input:

[15, 30, 20]

Output:

[15, 20]

Test case 2 :

Input:

[10, 20, 15, 10]

Output:

[10, 20, 15, 10]

Test case 3 :

Input:

[50, 15, 10, 20, 15]

Output:

[50, 15]

Test case 4 :

Input:

[15, 15, 15, 15, 15]

Output:

[15, 15, 15, 15, 15]

Test case 5 :

Input:

[30, 20, 15, 10]

Output:

[30, 15, 10]

Test case 6 :

Input:

[40, 25, 10]

Output:

[40, 10]

Q6. Fish Tank Filter Problem

A row of fish are swimming inside a long tank.

Each fish has a **size**.

The tank filter checks every **3 fishes that are next to each other**.

If the **total size** of those 3 fishes is **more than 60**, the **middle fish** swims away to make space.

This checking continues again and again until **no group of 3 fishes** has a size more than 60.

Your task is to return the **final list of fish sizes** after the tank becomes safe.

Rules :

- Look at every group of **3 fishes**.
- If their total size is **more than 60**, remove the **middle fish**.
- Keep repeating until no more fishes need to be removed.
- Finally, return the safe list of fish sizes.

Input Format:

A list of numbers showing the size of each fish.

Output Format:

A list of fish sizes after the tank becomes safe.

Sample Test Case 1:**Input:**

[20, 30, 15]

Output:

[20, 15]

Sample Test Case 2:**Input:**

[10, 50, 25, 20]

Output:

[10, 25, 20]

ANSWER:

Test case 1 :

Input:

[20, 30, 15]

Output:

[20, 15]

Test case 2 :

Input:

[10, 20, 15, 10]

Output:

[10, 20, 15, 10]

Test case 3 :

Input:

[50, 15, 10, 20, 15]

Output:

[50, 15]

Test case 4 :

Input:

[15, 15, 15, 15, 15]

Output:

[15, 15, 15, 15, 15]

Test case 5 :

Input:
[30, 20, 15, 10]

Output:
[30, 15, 10]

Test case 6 :

Input:
[40, 25, 10]
Output:
[40, 10]

Q7. Educational Credit Counter

A university registrar is processing student transcripts, logging the **Student ID**, the **Course Name**, and the **Credits Earned**. Some foreign transfer courses have credit values noted incorrectly (e.g., "Pass" instead of credits 3).

You must:

- Calculate the **total Credits Earned** for each **Student ID**.
- Invalid credit entries (non-integers) must be skipped.
- Print the final total credits dictionary and display the message **"Skipping invalid credit value"** for each skipped record.

Input Format:

A list of tuples with values in form: (Student ID, Course Name, Credits Earned)

Output Format:

Dictionary format {Student ID : Total Credits}

Sample Input 1 :

```
[('S001', 'Math', 4), ('S002', 'Sci', 3), ('S001', 'Eng', 'three'), ('S002', 'Hist', 3)]
```

Sample Output 1 :

```
Skipping invalid credit value  
{'S001': 4, 'S002': 6}
```

Sample Input 2:

```
[('S001', 'Math', 4), ('S002', 'Sci', 3), ('S002', 'Hist', 4)]
```

Sample Output 2 :

```
{'S001': 4, 'S002': 7}
```

ANSWER:

Test case 1 :

Input:

[('S001', 'Math', 4), ('S002', 'Sci', 3), ('S001', 'Eng', 'three'), ('S002', 'Hist', 3)]

Output:

Skipping invalid credit value

{ 'S001': 4, 'S002': 6 }

Test case 2 :

Input:

[('A', 'S', 'x')]

Output:

Skipping invalid credit value

{ }

Test case 3 :

Input:

[('A', 'S', 1)]

Output:

{ 'A': 1 }

Test case 4 :

Input:

[('A', 'S', 5), ('A', 'D', 5)]

Output:

{ 'A': 10 }

Test case 5 :

Input:

[('A', 'S', 5), ('B', 'S', 5)]

Output:

{ 'A': 5, 'B': 5 }

Test case 6 :

Input:

[('A', 'S', 5), ('A', 'D', 15)]

Output:

{ 'A': 20 }

Q8. Sports Player Game Stats

A sports team manager tracks player statistics during a season. Each record includes the **Player Name**, the **Game Type**, and the **Points Scored**. Some scores were incorrectly logged as fractions or text.

You must:

- Determine the **Total Points Scored** for each **Player Name**.
- **Only integer scores are valid**; For non-integer score, print a message **"Invalid record"**

skipped" for this.

- Print the final statistics with the key and value separated by a colon, all within a dictionary structure.

Input Format:

A list of tuples (Player Name, Game Type, Points Scored)

Output Format:

Dictionary {Player Name : Total Points}

Sample Input 1 :

```
[('Mayank', 'R', 10), ('Rohit', 'B', 5), ('Mayank', 'B', 15)]
```

Sample Output 1 :

```
{'Mayank': 25, 'Rohit': 5}
```

Sample Input 2:

```
[('Mia', 'R', 10), ('Zoe', 'B', 5), ('Mia', 'B', 15), ('Zoe', 'R', 'five')]
```

Sample Output 2:

```
Invalid record skipped  
{'Mia': 25, 'Zoe': 5}
```

ANSWER:

Test case 1 :

Input:

```
[('Mia', 'R', 10), ('Zoe', 'B', 5), ('Mia', 'B', 15), ('Zoe', 'R', 'five')]
```

Output:

```
Invalid record skipped  
{'Mia': 25, 'Zoe': 5}
```

Test case 2 :

Input:

```
[('A','S','x')]
```

Output:

```
Invalid record skipped  
{}
```

Test case 3 :

Input:

```
[('A','S',1)]
```

Output:

```
{'A': 1}
```

Test case 4 :

Input:

```
[('A','S',5),('A','D',5)]
```

Output:

```
{'A': 10}
```

Test case 5 :

Input:

```
[('A','S',5),('B','S',5)]
```

Output:

```
{'A': 5, 'B': 5}
```

Test case 6 :

Input:

```
[('A','S',5),('A','D',15)]
```

Output:

```
{'A': 20}
```

Q9. Construction Worker Hours Log

A construction foreman logs worker activities, recording the Worker Name, the Task Performed, and the Hours Worked on that specific task. The time clock occasionally logs faulty data.

You must:

- Calculate the total cumulative hours worked for each Worker Name.
- If the Hours Worked entry is not an integer, the entry is invalid and must be ignored.
- Display a message "Skipped Non-numeric Time Entry" for every invalid record.

Input Format:

A list of tuples: (Worker Name, Task Performed, Hours Worked)

Output Format:

Dictionary {Worker Name : Total Hours Worked}

Sample Input 1:

```
[('Raj', 'Weld', 5), ('Rohit', 'Paint', 8)]
```

Sample Output 1 :

```
{'Raj': 5, 'Rohit': 8}
```

Sample Input 2 :

```
[('Tom', 'Weld', 4), ('Jerry', 'Paint', 6), ('Tom', 'Drill', 'four')]
```

Sample Output 2 :

```
Skipped Non-numeric Time Entry  
{'Tom': 4, 'Jerry': 6}
```

ANSWER:Test case 1 :

Input:

```
[('Tom', 'Weld', 4), ('Jerry', 'Paint', 6), ('Tom', 'Drill', 'four')]
```

Output:

```
Skipped Non-numeric Time Entry  
{'Tom': 4, 'Jerry': 6}
```

Test case 2 :

Input:

```
[('A','S','x')]
```

Output:

```
Skipped Non-numeric Time Entry  
{}
```

Test case 3 :

Input:

```
[('A','S',1)]
```

Output:

```
{'A': 1}
```

Test case 4 :

Input:

```
[('A','S',5),('A','D',5)]
```

Output:

```
{'A': 10}
```

Test case 5 :

Input:

```
[('A','S',5),('B','S',5)]
```

Output:

```
{'A': 5, 'B': 5}
```

Test case 6 :

Input:

```
[('A','S',5),('A','D',15)]
```

Output:

```
{'A': 20}
```